Choose Your Own Adventure: An Applied Machine Learning Project

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PROJECT OVERVIEW

> This project is organized like a *Choose Your Own Adventure* story.[1] Teams select one of three paths, each with a different degree of independence and a different maximum grade. Regardless of the path, the objective is the same: use data and machine learning to answer a relevant question, evaluate the evidence carefully, and explain what a decision-maker should do with the results. The highest predictive score is not necessarily the best policy or practical recommendation.

Work in teams of **no more than three students**. Each member is expected to understand and defend the analysis. Creativity will be highly rewarded, but the final grade also depends on rigor, interpretability, reproducibility, and the relevance of the recommendation.

I. Choosing a path

Each team must select one of the following alternatives:

Path 1. Define your own problem and assemble your own data *Maximum possible grade: 10/10*

Identify an interesting question, explain why answering it matters, and obtain the data needed to investigate it. Your project should produce an insight that would not be obvious without the analysis. This path offers the greatest scope for originality and therefore the highest possible grade.

Path 2. Work with the professor's competition *Maximum possible grade: 9/10*

Use the competition and dataset available in this link. Your team is responsible for exploring the data, participating in the competition, developing and comparing models, studying their errors, and drawing a substantive conclusion. The challenge is not just to climb the leaderboard: it is to explain what the predictions imply for a decision.

Path 3. Revisit an established public competition *Maximum possible grade: 8/10*

Choose a well-known public competition (for example, one hosted on Kaggle or DrivenData) and use its data. Because these settings frequently have public solutions, tutorials, and substantial prior work, the maximum grade is lower. Identify and cite relevant previous work, distinguish your own contribution from what is already available, and provide an original, decision-oriented analysis rather than merely replicating a notebook.

I'm particularly interested in the story that can be told with the data. A great work has a compelling question, thoughtful feature engineering, careful attention to error costs, an innovative but defensible method, or an unusually useful recommendation. Model complexity by itself is not a contribution.

[1] For the curious, the link explains the reference

II. Steps

A. Initial meeting: before Week 4 (30%)

Every team must meet with the professor before the start of Week 4. Schedule the appointment through https://ignaciomsarmiento.github.io/. The meeting accounts for **30% of the project grade**.

Come to the initial meeting with a short slide presentation showing your research question, the data already in hand, initial exploratory results, and a credible plan for the remaining analysis. Be prepared to defend your choices.

B. Final presentation and submission (70%)

The last class will be dedicated to the final presentations. The team must submit its slides, the code underlying its results, and a short README with reproduction instructions to ignaciomsarmiento@gmail.com. This stage contributes the remaining **70% of the project grade**.

The presentation should be organized around *answering a relevant question*, not around displaying a succession of algorithms. Begin with a meaningful question and conclude with a clear answer, its limitations, and the implications for a relevant decision-maker: a policymaker, government agency, NGO, firm, or other stakeholder.

C. Minimum methodological standards

- Develop and evaluate machine learning models appropriate to your research question. Compare them against a credible baseline and well-motivated alternatives, using a common evaluation framework.
- Make the principal comparison under a **common evaluation design**: the same observations, predictors, splits, metrics, and comparable preprocessing. Isolate changes in features or preprocessing as separate experiments.
- Select metrics that reflect the substantive decision. Where possible, investigate uncertainty, score stability, thresholds, and relevant subgroups.
- Supply **reproducible code**, dependencies, an entry point, and instructions for obtaining or placing the data. Cite any borrowed solutions, code, and ideas. Do not distribute data without authorization.

III. Assessment

Let M and F denote the checkpoint and final-stage grades, each measured on a 0–10 scale. The project grade is

$$G_{\text{project}} = \min\{0.30M + 0.70F,\ C\}, \qquad C \in \{10, 9, 8\},$$

where C is the maximum grade corresponding to the chosen path.

Table 1: Evaluation criteria and weights

Checkpoint meeting	**Points**
Question, relevance, and intended decision	7
Data provenance, adequacy, and feasibility	8
Initial exploration and proposed machine learning strategy	10
Presentation and ability to answer questions	5
Checkpoint subtotal	*30*
Final presentation and reproducible analysis	**Points**
Machine learning methodology, validation, and model comparison	25
Answer to the research question and substantive implications	20
Originality, creativity, and insights	15
Clarity of presentation, defense, and reproducibility	10
Final-stage subtotal	*70*
Total	**100**

Guiding principle. A good machine learning project is not simply a collection of models: it offers a defensible answer to a meaningful question and a well-supported recommendation about what should be done next.